

ANSWERS

1.

Question	Answer	Marks
3(a)(i)	<u>mercury</u> exerts a changing or reduced pressure / force (when vertical but not when horizontal) / <u>mercury</u> stops exerting a pressure	B1
3(a)(ii)	particles hit walls / mercury	B1
	fewer hits per second or hit less often / less frequently or fewer hits per unit area	B1
	less force (per unit area on wall)	B1
3(b)	$(P =) dgh$ in any form	C1
	41 000 seen	C1
	140 000 (Pa)	A1
3(c)	curved shape with decreasing gradient	B1
	points plotted at or curve through $(\frac{1}{2}V_0, 2P_0)$, (P_0, V_0) and $(2V_0, \frac{1}{2}P_0)$	B1

2.

Question	Answer	Marks
3(a)	force divided by area	B1
3(b)	(molecules) move (around at random) / have K.E.	B1
	(molecules) hit sides / piston	B1
	(molecules) cause force on piston / sides	B1
3(c)(i)	$10 / 1.2 \times 10^{-4}$ or 8.3×10^4 (Pa) seen	C1
	1.8×10^5 Pa	A1
3(c)(ii)	$P_1 V_1 = P_2 V_2$ algebraic or numerical in any form or $P \propto 1/V$	B1
	27 or 28 cm ³	B1

3.

Question	Answer	Marks
9(a)(i)	molecules with similar / slightly smaller packing density and most molecules touching	B1
	random arrangement	B1
9(a)(ii)	molecules (of gas) further apart	B1
	(repulsive) forces between molecules (much) smaller	B1

4.

8(c)(i)	pressure increases with depth	B1
	pressure / force on outer face greater than that on inner face	B1
	resultant force on piston	B1
8(c)(ii)	pressure increases and more collisions (of molecules with walls of cylinder)	B1
	more frequent collisions or more collisions per unit area	B1
	molecules have less distance to travel between collisions (with walls) or molecular density greater	B1
8(c)(iii)	<u>curve</u> with negative gradient and gradient of decreasing magnitude	B1
	<u>negative gradient</u> at V_0 and approaches x-axis asymptotically	B1

5.

Question	Answer	Marks
5(a)	pressure = force / area	B1
	sharp knife has small(er) area and large(r) pressure or blunt knife has large(r) area and small(er) pressure	B1
5(b)(i)	temperature	B1
	mass of gas or number of molecules	B1
5(b)(ii)	molecules hit sides of container	B1
	twice as many molecules per second hit (unit area) or molecules hit twice as frequently	B1

6.

Question	Answer	Marks
4(a)	no / small(er) force between gas molecules (so they separate) or large(r) force between liquid molecules (keeps them together)	B1
	small(er) distance between liquid molecules or large(r) distance between gas molecules or density of liquid larger	B1
4(b)(i)	$P_1 V_1 = P_2 V_2$	B1
4(b)(ii)	volume change = 8×5 or 40 cm^3 or final volume = $60 \text{ (cm}^3\text{)}$ seen	C1
	$1.2 \times 10^5 \times 100 = p \times 60$	C1
	$2(.0) \times 10^5 \text{ Pa}$	A1
4(b)(iii)	more hits per second or hit more often / more frequently or more hits <u>on walls</u>	B1

- 7.
- (a) irregular arrangement of at least 8 molecules with at least one molecule touching other **B1**
- (b) solid – vibrate (about fixed positions) **B1**
 liquid – change position / slide (over each other) **B1**
or move / translate throughout (liquid)
or move in clusters
- gas – random movement **B1**
or move in all directions
or high speed / kinetic energy **or** have range of speeds
or move throughout container
or move in a straight line (between collisions)
or move freely
- (c) no / weak force between molecules in gases **B1**
or molecules not held together in gases
- speed / K.E of gas molecules fast(er) than solids **B1**
- 8.
- (a) 1. no fixed position/clusters/arranged randomly/close together/closely packed **B1**
 move throughout/at random/slide past each other/not in an organised way **B1**
2. regular/orderly arrangement/crystal lattice **or** fixed position **or** close together
 (ign. evenly spaced) **B1**
 vibrate **B1**
- (b) nothing/no change **and** increases **B1**
- 9.
- (a) solid more regular/ordered etc. or less space/separation between molecules or vv or solid molecules fixed and liquid molecules move throughout **B1**
- (b) (i) solids: strong(er) forces/bonds or energy not enough to break molecules free or vv **B1**
- (ii) fast(er)/high(er kinetic) energy molecules escape/evaporate **B1**
 molecules left are slower/less kinetic energy (on average) **B1**
- (iii) (hotter) molecules move faster/higher energy **B1**
 more molecules have energy/speed to break bonds/overcome forces **B1**